Ramsey theory: $R(3,3) = 6$, Ramsey's theorem, the binomial bound, $R(4,4) = 18$

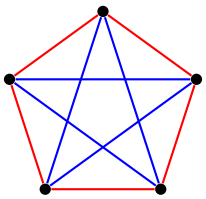
Notes from Carlos' lessons

Source lessons (chronological): 02-27, 03-01, 03-06, 04-10 (and alt), 04-16 (and alt). Code: `../code/ramsey-4-4.py`.

Setup. Colour the edges of the complete graph K_N with two colours. $R(r,s)$ is the least N such that every colouring has a red K_r or a blue K_s. (The 02-27 board uses green/purple and writes $R(n,m)$.) Immediate facts: $R(r,s) = R(s,r)$ (swap the colours), and $R(r,2) = r$ (either some edge is blue, or all edges are red and form a red K_r).

1 $R(3,3) = 6$

Lower bound, $R(3,3) > 5$. Colour K_5 with a red 5-cycle (the outer pentagon) and a blue 5-cycle (the pentagram). Neither colour contains a triangle. This is the 5-vertex picture on the 02-27 and 03-01 boards.



Upper bound, $R(3,3) \le 6$. Take any colouring of K_6 and a vertex v. Its 5 edges have two colours, so at least 3 have the same colour, say red to a, b, c. If any of ab, bc, ac is red, it forms a red triangle with v. Otherwise abc is a blue triangle. □

The double-counting attempt on the 02-27 board (my reading). The board counts, for K_6, the purple edges P and green edges G via triangles. Each edge lies in 4 triangles, so $4P = \sum_{\triangle} P(\triangle)$ and $4G = \sum_{\triangle} G(\triangle)$, and $P(\triangle) + G(\triangle) = 3$ gives $4(P+G) = 3\binom{6}{3} = 60$, i.e. $P + G = 15$. If there were no monochromatic triangle, each triangle would have 1 or 2 purple edges, so $20 \le 4P \le 40$, i.e. $5 \le P, G \le 10$. That is consistent with $P + G = 15$, so this count alone does *not* give a contradiction; the pigeonhole argument above is the one that proves $R(3,3) \le 6$. (The handwriting is faint; the numbers 15, $20 \le 4P \le 40$, $5 \le P \le 10$ are legible.)

2 Ramsey's theorem: $R(r, s)$ exists

Theorem 1 (Ramsey)**.** *For all $r, s \geq 2$, $R(r, s)$ exists.*

Proof. Induction on $r + s$ (the board's hypothesis $Q(N)$). Base: $R(r, 2) = r$ and $R(2, s) = s$ exist; in particular $R(2, 2) = 2$ (the case $r + s = 4$).

Step: assume $R(r - 1, s)$ and $R(r, s - 1)$ exist. We claim $R(r, s) \leq R(r - 1, s) + R(r, s - 1)$.

Take a colouring of K_N with $N = R(r - 1, s) + R(r, s - 1)$ and a vertex v. Let R_v be its red neighbours and B_v its blue neighbours, so $|R_v| + |B_v| = N - 1 = R(r - 1, s) + R(r, s - 1) - 1$. If both $|R_v| \leq R(r - 1, s) - 1$ and $|B_v| \leq R(r, s - 1) - 1$, the sum would be at most $N - 2$. So either $|R_v| \geq R(r - 1, s)$ or $|B_v| \geq R(r, s - 1)$.

- *Case* $|R_v| \geq R(r - 1, s)$. Inside R_v there is a red K_{r-1} or a blue K_s. A red K_{r-1} together with v is a red K_r.
- *Case* $|B_v| \geq R(r, s - 1)$. Symmetric: a blue K_{s-1} together with v is a blue K_s, or B_v already holds a red K_r.

Either way the colouring contains a red K_r or a blue K_s. □

3 The explicit bound $R(r, s) \leq \binom{r+s-2}{r-1}$

The 03-01 board turns the inequality into the recurrence $U(r, 2) = r$, $U(2, s) = s$, $U(r, s) = U(r-1, s) + U(r, s - 1)$, so $R(r, s) \leq U(r, s)$ by induction. The 03-06 board solves it with Pascal's identity.

Theorem 2. $R(r, s) \leq \dbinom{r + s - 2}{r - 1}$.

Proof. Induction on $r + s$. Base: $\binom{r}{r-1} = r$ and $\binom{s}{1} = s$. Step:

$$R(r, s) \leq R(r - 1, s) + R(r, s - 1) \leq \binom{r + s - 3}{r - 2} + \binom{r + s - 3}{r - 1} = \binom{r + s - 2}{r - 1}.$$

□

Sample consequences: $R(3, 3) \leq \binom{4}{2} = 6$ and $R(4, 4) \leq \binom{6}{3} = 20$ (the board also tabulates the binomials in a grid). It also lists $43 \leq R(5, 5) \leq 46$; further bounds for $R(6, 6)$ and $R(10, 10)$ are on that board, but the handwriting is too small for me to transcribe them reliably.

4 $R(4, 4) = 18$

Notation on the 04-10 board: "$n \to (b, r)$" means every colouring of K_n (blue/red) has a blue K_b or a red K_r; so $R(b, r)$ is the least such n. Known: $6 \to (3, 3)$ and $4 \to (2, 4)$.

4.1 Upper bound: $18 \to (4,4)$

Lemma 3 (Parity improvement)**.** *Let $n_1 \to (b, r-1)$ and $n_2 \to (b-1, r)$. Then $n_1 + n_2 \to (b, r)$; and if n_1, n_2 are both even, already $n_1 + n_2 - 1 \to (b, r)$.*

Proof. Take a colouring of K_N with $N = n_1 + n_2 - 1$ (the weaker claim with $N = n_1 + n_2$ is the argument of Section 2). Each vertex v has $N - 1 = n_1 + n_2 - 2$ neighbours.

- If $\deg_R(v) \geq n_1$: the red neighbourhood has $\geq n_1 \to (b, r-1)$ vertices, so it contains a blue K_b (done) or a red K_{r-1} (add v to get a red K_r).
- If $\deg_B(v) \geq n_2$: symmetrically, a blue K_{b-1} plus v is a blue K_b, or a red K_r appears.

So if the colouring has no blue K_b and no red K_r, no vertex satisfies either condition: $\deg_R(v) \leq n_1 - 1$ and $\deg_B(v) \leq n_2 - 1$ for all v. Their sum is $n_1 + n_2 - 2$, so both are equalities: $\deg_R(v) = n_1 - 1$ for every v. If n_1 is even, $n_1 - 1$ is **odd**, and $N = n_1 + n_2 - 1$ is odd. So the graph has an odd number of vertices, each with odd red degree, and $\sum_v \deg_R(v)$ is odd. But by the handshake lemma it equals $2|E_R|$, which is even. Contradiction. □

Applications.

- $R(3,4) \leq 9$: take $b = 3$, $r = 4$, $n_1 = R(3,3) = 6$, $n_2 = R(2,4) = 4$, both even, so $6 + 4 - 1 = 9$. (Known: $R(3,4) = 9$.)
- $R(4,4) \leq R(4,3) + R(3,4) = 9 + 9 = 18$.

4.2 Lower bound: $17 \not\to (4,4)$ (the Paley graph on $\mathbb{Z}_{17}$)

Vertices $0, 1, \ldots, 16$. Let $Q = \{x^2 \bmod 17\} = \{1, 2, 4, 8, 9, 13, 15, 16\}$ be the non-zero quadratic residues (the squares $1, 4, 9, 16, 25 \equiv 8, 36 \equiv 2, 49 \equiv 15, 64 \equiv 13$). Colour $\{x, y\}$ red if $x - y \in Q$ and blue otherwise. This is well defined because $-1 \equiv 4^2 \in Q$ (the board checks $x > y$ and $x - y \in \{1, 2, 4, 8, 9, 13, 15, 16\}$). Each vertex has red degree 8 and blue degree 8.

Facts about Q. Take the primitive root $g = 3$, so $\mathbb{Z}_{17}^\times = \{g^0, \ldots, g^{15}\}$ and $g^{16} = 1$. Then Q is exactly the set of even powers of g. Indeed $(g^t)^2 = g^{2t}$ is even; conversely, if $g^m = (g^t)^2$ with m odd, then $g^{m-2t} = 1$ forces $16 \mid m - 2t$, impossible by parity. Hence, writing N for the non-residues (the odd powers):

- $Q \cdot Q = Q$, $N \cdot N = Q$, $Q \cdot N = N$, as in the board's multiplication table.
- **Symmetry.** Multiplying all vertices by $q \in Q$ maps red edges to red edges (since $q(x - y) \in Q \iff x - y \in Q$); multiplying by a non-residue swaps the colours. So the graph is isomorphic to its complement, and translations $x \mapsto x + r$ are also symmetries.

No red K_4. Suppose $\{a, b, c, d\}$ is a red K_4. Translate by $-a$ and scale by $(b - a)^{-1} \in Q$ (allowed because $b - a \in Q$): we may assume the clique is $\{0, 1, p, q\}$. Red edges to 0 and to 1 require p, $q \in Q$ and $p - 1, q - 1 \in Q$. Checking Q:

$p \in Q$	1	2	4	8	9	13	15	16
$p - 1 \in Q$?	no (0)	yes	no (3)	no (7)	yes	no (12)	no (14)	yes

So $p, q \in \{2, 9, 16\}$ (the board boxes exactly these three). But $2 - 9 \equiv 10$, $2 - 16 \equiv 3$, $9 - 16 \equiv 10$, and none is in Q, so no two of them are red-adjacent. Contradiction. (Also checked by brute force over all $\binom{17}{4}$ subsets.)

No blue K_4. Multiplication by a non-residue (say 3) maps a blue K_4 to a red K_4. □

So $R(4, 4) \geq 18$, and with the upper bound, $R(4, 4) = 18$.

Notes

- The "theory-builders vs. problem-solvers" remark in the top-right of the 03-06 board (Euler/Gauss/Grothendieck vs. Erdős) and the van der Waerden question beside it are asides; the latter is picked up in the topic *Van der Waerden's theorem*.
- The `-alt` images are slightly different exports of the 04-10 and 04-16 boards with the same content.
- Greenwood–Gleason (1955) first showed the 17-vertex lower bound; see Proving that $R(4, 4) > 17$ (Plus Magazine). The web sources I found did not spell out the parity argument for the upper bound, so the lemma above is my write-up of what the 04-10 board sketches (its labels: $n_1 \to (b, r - 1)$, $n_2 \to (b - 1, r)$, odd/even degrees, "$\sum \deg v = 2|E|$ even!").